SAMVED

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Rugged PCB-Based Product Development for Real-World Deployment

1. From Prototype to Product

The current system is built as a working prototype using development boards and exposed components. It proves the idea.

But let's be honest—what works on a table doesn't survive inside a manhole.

The next step is different. It’s not about proving the concept anymore. It’s about making it deployable, repeatable, and durable at scale.

That’s where PCB-based design and rugged product engineering come in.

This report focuses on how the system will evolve into:

- A mass-producible PCB-based device
- A wearable form factor (watch-like design)
- A rugged system capable of handling water, impact, and scratches

This builds directly on the current design approach described in your earlier report .

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2. Design Goals for Product Version

At this stage, the focus shifts from “working” to “lasting”.

The product must:

- Handle water exposure and humidity
- Resist scratches, drops, and rough usage
- Maintain stable sensor performance
- Be compact and wearable
- Be easy to manufacture using PCB assembly

In short, it should behave less like a project—and more like a field tool.

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3. Transition to PCB-Based Design

Why Move to PCB?

Using ready-made boards like ESP32 modules is great for prototyping. But for production:

- They are bulky
- Connections are exposed
- Reliability is inconsistent

A custom PCB solves this.

It allows:

- Compact layout
- Stronger connections
- Better protection options
- Easier mass production

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4. PCB Design Changes for Ruggedness

This is the core of your upgrade.

4.1 Compact and Layered PCB Layout

Instead of spreading components loosely:

- Use a compact PCB layout
- Place critical components (MCU, power) centrally
- Keep sensor interfaces at the edge

This reduces exposure and improves structural strength.

4.2 Conformal Coating (Standard for All Boards)

Every PCB in the product version will include:

- A thin protective coating layer
- Protection against:
    - Moisture
    - Corrosion
    - Dust

It’s like adding an invisible shield over the electronics.

4.3 Selective Potting for Critical Zones

Instead of covering everything:

- Apply silicone potting only in vulnerable sections:
    - Power circuits
    - Connector joints

Leave sensors unpotted.

This balance ensures:

- Protection where needed
- Functionality where required

4.4 Connector Reduction Strategy

Loose wires are failure points.

So the design will:

- Minimize external wiring
- Use direct PCB-mounted connectors
- Prefer flex cables or sealed connectors

Fewer connections = fewer failures.

4.5 Edge Protection and Mounting

PCB edges are often ignored—but they matter.

Proposed changes:

- Rounded PCB edges
- Proper mounting points with screws
- Rubber spacers inside enclosure

This prevents cracking during drops.

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5. Enclosure Design for Product Version

The enclosure now becomes part of the engineering—not just a box.

5.1 Water Resistance

- Use IP67-rated enclosure design
- Add:
    - Rubber gaskets
    - Tight sealing joints

For the wearable:

- Use watch-style sealed casing

5.2 Scratch Resistance

- Use polycarbonate or ABS plastic
- Add:
  - Matte or textured finish
  - Optional rubber outer layer

This prevents visible wear during daily use.

5.3 Sensor Exposure Design

Sensors need air—but not water.

Solution:

- Dedicated sensor chamber
- Covered with:
  - Metal mesh
  - Hydrophobic membrane

This allows airflow while blocking liquid entry.

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6. Wearable Design Upgrade (Watch Form Factor)

The wearable unit will evolve into something closer to a smartwatch-style device.

Key Changes

- Compact circular or rectangular PCB
- Wrist-mounted strap design
- Front-facing sensor vents
- Sealed buttons or touch interface

Internal Improvements

- Integrated buzzer and LED
- Small battery inside sealed compartment

- Reinforced casing to handle sweat, rain, and impact

This makes the device:

- Comfortable to wear
- Always active
- Hard to damage during work

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7. Probe Unit Design Improvements

The probe also needs refinement.

Structural Changes

- Strong cylindrical housing
- Internal PCB mounted securely
- Sensor section isolated in vented chamber

Cable Improvements

- Use industrial cable with outer insulation
- Add strain relief at entry point
- Use waterproof cable gland

Also:

- Load handled by rope—not cable

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8. Additional Protection Measures

To further improve durability:

- Add rubber bumpers on outer edges
- Use shock-absorbing mounts inside enclosure
- Design for easy maintenance and part replacement

These small decisions reduce long-term failure.

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9. Manufacturing Considerations

This design supports scaling.

- PCB can be mass-produced using standard fabrication
- Components are easily available
- Assembly can be done with basic production setup

No need for expensive industrial processes at early stage.

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10. Final Positioning

This phase does not claim a finished industrial product.

Instead, it presents:

- A clear pathway from prototype to deployable system
- Practical design upgrades for real-world conditions
- Thoughtful component and PCB-level improvements